

Supervised fine-tuning of decoder-only model for russian spellcheck task

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Abstract

My project is devoted to spellcheck problem. I tested if a decoder-based model can beat encoder-decoder models that was provided by SberDevices. During tests I found that there is no label if data was typed on desktop, mobile or tablet device. I've decided to create a dataset to solve this problem. To create this dataset I've created an app to help gather this data. The dataset and models were pushed to hugging face. Project repo:https://github.com/BorisVIkVik/SpellCheck_Experiments_Project.git. Dataset app repo:https://github.com/BorisVIkVik/SpellCheck_Game_Dataset.git.

1 Introduction

In modern time we type a lot of text, so spellcheck task becomes very important to save time or not to lose face. There are articles [Martynov et al., 2024] that study this task, but they don't take account of the device that person use to type. Typing on mobile devices is a huge part of modern data, but russian open datasets that are used for spellcheck task don't label the type of device. It is happening because of gathering data that was already in the internet and you don't have any way to decide from what device it was typed. To face this problem I've made an game like app where people get sentences that they need to type and the app checks the device type that they used.

1.1 Team

Viktorov Boris - one man army.

2 Related Work

[Martynov et al., 2024] is a SberDevices teams article where they proposed the SAGE framework and explored the suitability of the M2M approach for spell checking.

3 Model Description

Base model was Qwen3.5-0.8B model. I've used Unsloth to perform SFT. SFT type was Instruction tuning.

4 Dataset

For this task I've decided to create my dataset. For the purpose of gathering data there was developed an app for people to type the provided message from russian classic literature. The app saves original text, typed text, anonymous user id (in dataset it's called `player_id`), timestamp and device type. To get the device the app checks the device that user uses to type and gives this example one of three labels: desktop, mobile or tablet. The dataset¹ was pushed to hugging face and called `RU_SPELLCHECK_DEVICE`.

On the Tab. 1 you can see the statistics for the mentioned dataset.

	Train	Test
Sentences	269	269
Device types	3	
Different users	33	

Table 1: Statistics of the `RU_SPELLCHECK_DEVICE`.

5 Experiments

This section should include several subsections.

5.1 Metrics

I've used Errant based algorithm from sage framework to calculate the metrics. Algorithms output Precision, Recall and F1 scores that can be interpreted like the following:

Precision: one minus share of unnecessary amendments; Recall: proportion of expected corrections; F1: famous geometric mean of aforementioned two;

5.2 Experiment Setup

Training was done in google colab.

¹ BW/RU_SPELLCHECK_DEVICE.

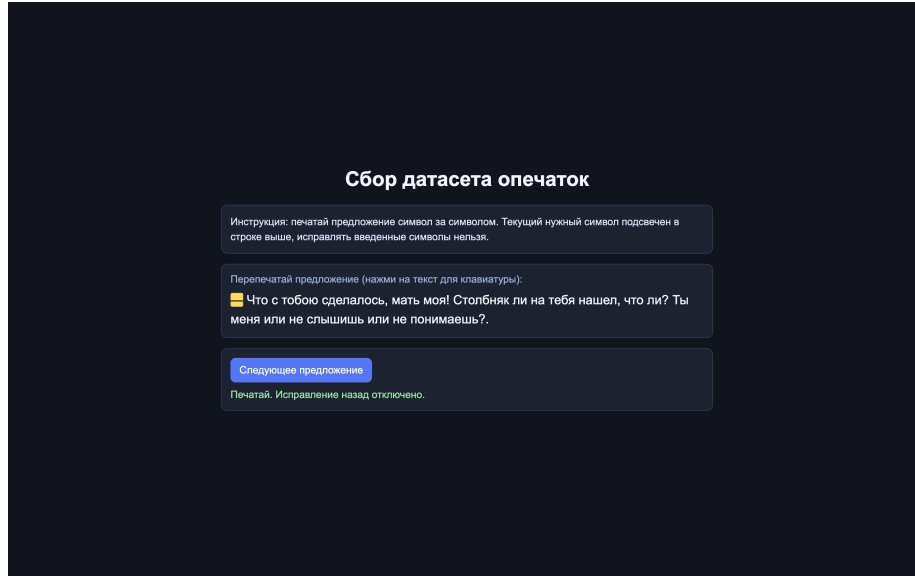


Figure 1: The app for dataset gathering.

5.3 Honorable mention

New dataset give interesting opportunities in text classification task, because we can try to predict the device that the person uses. This maybe useful for improving spellcheck because typing on different device causes different error. Also in maybe useful in Criminalistics. Here are results of simple test. I've fine-tuned DeepPavlov/rubert-base-cased for predicting device label using typed text.

Table 2: Экспериментальные метрики оценки модели (эпоха 5)

Метрика	Значение
Eval loss	0.73699
Eval accuracy	0.68919
Eval F1 score	0.68411
Eval precision	0.69390
Eval recall	0.68919
Eval runtime (s)	2.6592
Eval samples per second	111.313
Eval steps per second	7.145
Epoch	5

6 Results

The results of my model on test split of my dataset:

Table 3: Fine-tuned Qwen Results

Metric Type	Precision (%)	Recall (%)	F1 (%)
Overall	35.17	31.28	33.11
SPELL	39.60	37.02	38.27
CASE	47.73	50.00	48.84
PUNCT	53.85	50.30	52.01
YO	0.00	0.00	0.00

And here is for Sberdevices best model ai-forever/sage-fredt5-large:

Table 4: sage-fredt5-large

Metric Type	Precision (%)	Recall (%)	F1 (%)
Overall	64.42	73.73	68.76
SPELL	59.91	65.42	62.55
CASE	17.24	47.62	25.32
PUNCT	11.38	25.75	15.78
YO	0.00	0.00	0.00

7 Conclusion

I have collected dataset with device label that is unique opensource dataset because of this. I’ve trained decoder model that outperformed SberDevices models in punctuation and case metrics.

References

- [Martynov et al., 2024] Martynov, N., Baushenko, M., Kozlova, A., Kolomeytseva, K., Abramov, A., and Fenogenova, A. (2024). A methodology for generative spelling correction via natural spelling errors emulation across multiple domains and languages. In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 138–155.